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Laser device and leak check method for laser device

Abstract

A leak check method for a laser device includes exposing a closed space accommodating laser medium gas to the atmosphere, isolating the closed space from the atmosphere after exposing the closed space to the atmosphere, introducing neon-containing gas containing neon gas to the closed space, and determining whether or not the neon gas is leaking to outside of the closed space.

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Primary Examiner: Carter; Michael

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims the benefit of International Application No.

PCT/JP2019/039358, filed on Oct. 4, 2019, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a laser device and a leak check method for the laser device.

2. Related Art

(3) Recently, in a semiconductor exposure apparatus, improvement in resolution has been desired for miniaturization and high integration of semiconductor integrated circuits. For this purpose, an exposure light source that outputs light having a shorter wavelength has been developed. For example, as a gas laser device for exposure, a KrF excimer laser device for outputting laser light having a wavelength of about 248 nm and an ArF excimer laser device for outputting laser light having a wavelength of about 193 nm are used.

(4) The KrF excimer laser device and the ArF excimer laser device each have a large spectral line width of about 350 to 400 pm in natural oscillation light. Therefore, when a projection lens is formed of a material that transmits ultraviolet rays such as KrF laser light and ArF laser light, there is a case in which chromatic aberration occurs. As a result, the resolution may decrease. Then, a spectral line width of laser light output from the gas laser device needs to be narrowed to the extent that the chromatic aberration can be ignored. For this purpose, there is a case in which a line narrowing module (LNM) including a line narrowing element (etalon, grating, and the like) is provided in a laser resonator of the gas laser device to narrow a spectral line width. In the following, a gas laser device with a narrowed spectral line width is referred to as a line narrowing gas laser device.

LIST OF DOCUMENTS

Patent Documents

(5) Patent Document 1: Japanese Unexamined Utility Model Application Publication No. H01-

146565 Patent Document 2: U.S. Pat. No. 6,819,699 Patent Document 3: Japanese Patent

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SUMMARY

(6) A leak check method for a laser device according to an aspect of the present disclosure includes exposing a closed space accommodating laser medium gas to the atmosphere, isolating the closed space from the atmosphere after exposing the closed space to the atmosphere, introducing neon-containing gas containing neon gas to the closed space, and determining whether or not the neon gas is leaking to outside of the closed space.

(7) A leak check method for a laser device according to another aspect of the present disclosure is a leak check method for a laser device including a laser chamber, a first pipe configured to introduce first gas containing neon gas and fluorine gas into the laser chamber, and a second pipe configured to introduce second gas containing neon gas and having fluorine gas concentration lower than that of the first gas into the laser chamber. Here, the leak check method includes introducing third gas containing neon gas and having fluorine gas concentration lower than that of the first gas into a closed space including a space inside the laser chamber, and determining whether or not neon gas is leaking to outside of the closed space.

(8) A laser device according to an aspect of the present disclosure includes a laser chamber, a first pipe configured to introduce first gas containing neon gas and fluorine gas into the laser chamber, a second pipe configured to introduce second gas containing neon gas and having fluorine gas concentration lower than that of the first gas into the laser chamber, and a third pipe configured to introduce third gas containing neon gas and having fluorine gas concentration lower than that of the first gas into the laser chamber and the first pipe.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the present disclosure will be described below merely as examples with reference to the accompanying drawings.
- (2) FIG. 1 schematically shows the configuration of a laser device according to a comparative example.
- (3) FIG. 2 is a flowchart showing the procedure of a leak check method in a comparative example.
- (4) FIG. 3 is a diagram for explaining the range of a closed space.
- (5) FIG. 4 is a flowchart showing the procedure of maintenance operation in the comparative example.
- (6) FIG. 5 is a flowchart showing the procedure of leak check in the comparative example.
- (7) FIG. 6 schematically shows the configuration of the laser device according to a first embodiment.
- (8) FIG. 7 is a flowchart showing the procedure of the maintenance operation in the first embodiment.
- (9) FIG. 8 is a flowchart showing the procedure of the leak check in the first embodiment.
- (10) FIG. 9 is a flowchart showing the procedure of the leak check in a second embodiment.
- (11) FIG. 10 schematically shows the configuration of a replacement laser chamber.
- (12) FIG. 11 is a flowchart showing the procedure of the leak check in a third embodiment.

DESCRIPTION OF EMBODIMENTS

(13) Content

(14) 1. Laser Device According to Comparative Example 1.1 Configuration 1.1.1 Laser Oscillator and Various Control Units 1.1.2 Various Gas Supply Devices and Exhaust Device 1.1.2.1 Laser Gas Supply Device 1.1.2.2 Chamber Purge Gas Supply Device 1.1.2.3 Exhaust Device 1.1.2.4 Optical Module Purge Gas Supply Device 1.2 Operation 1.2.1 Operation of Laser Oscillator 1.2.2 Operation of Optical Module Purge Gas Supply Device 1.2.3 Leak Check Method 1.2.3.1 Maintenance Operation 1.2.3.2 Leak Check 1.3 Problems of Comparative Example 2. Laser Device that Performs Leak Check Using Neon-containing Gas 2.1 Configuration 2.2 Operation 2.2.1 Maintenance Operation 2.2.2 Leak Check 2.3 Effect 3. Laser Device that Performs Leak Check Based on Change in Gas Pressure 4. Laser Device that Performs Leak Check of Replacement Laser Chamber 4.1 Replacement Laser Chamber 4.2 Leak Check Procedure 4.3 Effect 5. Others

(15) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings. The embodiments described below show some examples of the present disclosure and do not limit the contents of the present disclosure. Also, all configurations and operation described in the embodiments are not necessarily essential as configurations and operation of the present disclosure. Here, the same components are denoted by the same reference numerals, and duplicate description thereof is omitted.

(16) 1. Laser Device According to Comparative Example

(17) 1.1 Configuration

(18) FIG. 1 schematically shows the configuration of a laser device **1** according to a comparative example. The laser device **1** is used together with an exposure apparatus **100**.

(19) The laser device **1** includes a laser chamber **10**, a line narrowing module **14**, an output coupling mirror **15**, a laser control unit **30**, and a gas control unit **32**. The line narrowing module **14** and the output coupling mirror **15** configure a laser resonator. The laser device **1** further includes a laser gas supply device **33**, a chamber purge gas supply device **35**, an exhaust device **37**, and an optical module purge gas supply device **39**.

(20) 1.1.1 Laser Oscillator and Various Control Units

(21) The laser chamber **10** is arranged in the optical path of the laser resonator. The laser chamber **10**, the laser resonator, and a high-voltage power source (not shown) constitute a laser oscillator.

(22) The laser chamber **10** is provided with two windows **10a**, **10b**. The laser chamber **10** accommodates a pair of electrodes **11a**, **11b**, and further a laser medium gas. The laser medium gas includes halogen gas, rare gas, and buffer gas. For example, the halogen gas is fluorine gas, the rare gas is argon gas or krypton gas, and the buffer gas is neon gas. Further, the laser medium gas may include xenon gas as additive gas. The laser chamber **10** is connected to a pressure gauge **P**.

(23) The line narrowing module **14** includes wavelength selection elements such as a prism **14a** and a grating **14b**. The line narrowing module **14** is accommodated in a housing **18**. The output coupling mirror **15** is configured by a partial reflection mirror. The output coupling mirror **15** is accommodated in a housing **19**. The housing **19** is provided with a window **10c**. The housing **19** may further accommodate an energy monitor (not shown) for measuring the pulse energy of pulse laser light.

(24) The laser control unit **30** is configured by a computer unit including a central processing unit (CPU) **30a** and a memory **30b**. The memory **30b** stores programs and data necessary for information processing. The CPU **30a** is configured to read out various kinds of data and perform information processing in accordance with the programs stored in the memory **30b**. The laser control unit **30** is connected to an exposure apparatus control unit (not shown) of the exposure apparatus **100**, a fault detection and classification (FDC) system **120**, and a user interface **140**.

(25) The gas control unit **32** is configured by a computer unit including a CPU **32a** and a memory **32b**. The memory **32b** stores programs and data necessary for information processing. The CPU **32a** is configured to read out various kinds of data and perform information processing in accordance with the programs stored in the memory **32b**.

(26) 1.1.2 Various Gas Supply Devices and Exhaust Device

(27) Next, the laser gas supply device **33**, the chamber purge gas supply device **35**, the exhaust device **37**, and the optical module purge gas supply device **39** will be described. In the following description, a first end of each pipe through which gas flows is the end on the upstream side of the gas flow, and a second end thereof is the end on the opposite side of the first end.

(28) 1.1.2.1 Laser Gas Supply Device

(29) The laser gas supply device **33** includes a valve **F2-V1** arranged in the vicinity of a first end of a pipe **331** and a valve **B-V1** arranged in the vicinity of a first end of a pipe **332**. The pipe **331** corresponds to the first pipe in the present disclosure. The pipe **332** corresponds to the second pipe in the present disclosure.

(30) A first end of the pipe **331** is connected to a fluorine-containing gas cylinder **23** via a pipe **231**. The fluorine-containing gas cylinder **23** corresponds to the first gas supply source in the present disclosure. The fluorine-containing gas cylinder **23** stores fluorine-containing gas containing neon gas and fluorine gas. The fluorine-containing gas corresponds to the first gas in the present disclosure. The fluorine-containing gas is, for example, mixed gas of fluorine gas, argon gas, and neon gas.

(31) The first end of the pipe **332** is connected to a neon-containing gas cylinder **24** via a pipe **241**. The neon-containing gas cylinder **24** corresponds to the second gas supply source in the present disclosure. The neon-containing gas cylinder **24** stores neon-containing gas containing neon gas and having fluorine gas concentration lower than that of the fluorine-containing gas. The neon-containing gas corresponds to the second gas in the present disclosure. The neon-containing gas

may not contain fluorine gas. The neon-containing gas is, for example, mixed gas of argon gas, neon gas, and xenon gas.

(32) Alternatively, the fluorine-containing gas may be mixed gas of fluorine gas, krypton gas, and neon gas, and the neon-containing gas may be mixed gas of krypton gas and neon gas.

(33) A second end of the pipe **331** and a second end of the pipe **332** are connected to each other and are further connected to a pipe **333**. The pipe **333** is connected to a pipe **334**, and the pipe **334** is connected to the laser chamber **10**. A check valve **332a** for suppressing backflow of the fluorine-containing gas from the pipe **331** and the pipe **333** is arranged in the vicinity of the second end of the pipe **332**. Each of the pipes **331**, **332** may be provided with a mass flow controller (not shown).

(34) 1.1.2.2 Chamber Purge Gas Supply Device

(35) The chamber purge gas supply device **35** includes a valve L-V1 arranged in the vicinity of a first end of a pipe **351**. The first end of the pipe **351** is connected to a helium gas cylinder **25** via a pipe **251**. A second end of the pipe **351** is connected to a connection portion between the pipe **333** and the pipe **334**. A check valve **351a** for suppressing backflow of the fluorine-containing gas and the neon-containing gas from the pipe **333** and the pipe **334** is arranged in the vicinity of the second end of the pipe **351**.

(36) A first end of a pipe **352** is connected to the pipe **351** between the valve L-V1 and the check valve **351a**. A second end of the pipe **352** is connected to the pipe **331**. A check valve **352a** for suppressing backflow of the fluorine-containing gas from the pipe **331** is arranged in the vicinity of the second end of the pipe **352**.

(37) A first end of a pipe **353** is connected to the pipe **351** between the valve L-V1 and the check valve **351a**. A second end of the pipe **353** is connected to the pipe **332**. A check valve **353a** for suppressing backflow of the neon-containing gas from the pipe **332** is arranged in the vicinity of the second end of the pipe **353**.

(38) 1.1.2.3 Exhaust Device

(39) The exhaust device **37** includes a fluorine trap **37a** and an exhaust pump **37b** arranged at a pipe **373**. The fluorine trap **37a** contains a processing agent for capturing fluorine gas and fluorine compounds. The exhaust pump **37b** is configured to forcibly exhaust gas having passed through the fluorine trap **37a**. The pipe **373** may include a bypass flow path (not shown) that bypasses the exhaust pump **37b** to discharge the gas having passed through the fluorine trap **37a** when the gas pressure in the laser chamber **10** is higher than atmospheric pressure.

(40) A first end of the pipe **373** is connected to a second end of a pipe **371** and a second end of a pipe **372**. The first end of the pipe **371** is connected to the laser chamber **10**. A valve EX-V1 is arranged at the pipe **371**. A first end of the pipe **372** is connected to a connection portion between the pipe **331** and the pipe **332**. A valve EX-V2 is arranged at the pipe **372**.

(41) 1.1.2.4 Optical Module Purge Gas Supply Device

(42) The optical module purge gas supply device **39** includes a valve N2-V1 arranged in the vicinity of a first end of a pipe **391**. The first end of the pipe **391** is connected to a nitrogen gas cylinder **29** via a pipe **291**. A second end of the pipe **391** is branched into a pipe **392** and a pipe **393**. The pipe **392** is connected to the housing **18** that accommodates the line narrowing module **14**. The pipe **393** is connected to the housing **19** that accommodates the output coupling mirror **15**. Each of the pipes **392**, **393** may be provided with a mass flow controller (not shown).

(43) 1.2 Operation

(44) 1.2.1 Operation of Laser Oscillator

(45) The laser control unit **30** transmits a trigger signal to a high-voltage power source (not shown) in accordance with a control signal from the exposure apparatus control unit. The high-voltage power source generates a pulse high voltage in accordance with the trigger signal, and applies the high voltage between the electrodes **11a**, **11b**.

(46) When the high voltage is applied between the electrodes **11a**, **11b**, discharge occurs between the electrodes **11a**, **11b**. The laser medium gas in the laser chamber **10** is excited by the energy of

the discharge and shifts to a high energy level. When the excited laser medium gas then shifts to a low energy level, light having a wavelength corresponding to the difference between the energy levels is emitted.

(47) The light generated in the laser chamber **10** is output to the outside of the laser chamber **10** through the windows **10a**, **10b**. The light output from the window **10a** is line-narrowed by the line narrowing module **14**, and light having wavelength close to the desired wavelength is returned to the laser chamber **10** through the window **10a**.

(48) The output coupling mirror **15** transmits and outputs a part of the light output from the window **10b**, and reflects the other parts back into the laser chamber **10** through the window **10b**.

(49) In this way, the light output from the laser chamber **10** reciprocates between the line narrowing module **14** and the output coupling mirror **15**. This light is amplified every time it passes through a discharge space between the discharge electrodes **11a**, **11b**. Further, the light is line-narrowed each time being turned back in the line narrowing module **14**. Thus, the light having undergone laser oscillation and line narrowing is output as pulse laser light from the laser resonator through the output coupling mirror **15** and output from the laser device **1** through the window **10c**.

(50) Pulse laser light output from the laser device **1** is incident on the exposure apparatus **100**.

(51) 1.2.2 Operation of Optical Module Purge Gas Supply Device

(52) When the laser device **1** is in operation, the optical module purge gas supply device **39** introduces nitrogen gas into the housings **18**, **19**. At this time, the valve N2-V1 is opened, and the valves F2-V1, B-V1, L-V1, EX-V1, EX-V2 are closed.

(53) The introduced amount of the nitrogen gas may be controlled by the mass flow controller (not shown) arranged at each of the pipes **392**, **393**. The nitrogen gas stays in the housings **18**, **19**, and is then exhausted from exhaust ports **18a**, **19a**. By introducing the nitrogen gas into the housings **18**, **19**, contamination of the optical elements in the housings **18**, **19** is suppressed.

(54) 1.2.3 Leak Check Method

(55) FIG. **2** is a flowchart showing the procedure of a leak check method in the comparative example. The leak check method is performed in the order of maintenance operation (S10) and leak check (S30).

(56) The maintenance operation (S10) includes exposing a closed space accommodating the laser medium gas to the atmosphere, then performing component replacement of the laser chamber **10**, and then isolating the closed space from the atmosphere.

(57) The leak check (S30) includes introducing helium gas into the closed space isolated from the atmosphere, and then determining whether or not the helium gas is leaking to the outside of the closed space.

(58) FIG. **3** is a diagram for explaining the range of the closed space. FIG. **3** shows the laser chamber **10** and various pipes in the laser device **1** shown in FIG. **1**, and shows the range of the laser chamber **10** and the pipes, which form the closed space, by broken lines. The closed space includes a part of the space inside each of the pipes **331**, **332**, **351**, **371**, **372**, the space inside each of the pipes **333**, **334**, **352**, **353**, and the space inside the laser chamber **10**.

(59) 1.2.3.1 Maintenance Operation

(60) FIG. **4** is a flowchart showing the procedure of the maintenance operation in the comparative example. The flowchart of FIG. **4** corresponds to the subroutine of S10 in FIG. **2**. In the procedure shown in FIG. **4**, steps each surrounded by a solid line frame indicate operation of the gas control unit **32**, and steps each surrounded by a broken line frame indicate operation of an operator.

(61) In the maintenance of the laser chamber **10**, fluorine gas in the closed space is removed by the following process before the closed space is exposed to the atmosphere.

(62) In S11, the gas control unit **32** sets the value of a counter N for counting the number of repetitions of gas exhaust to an initial value 1.

(63) In S12, the gas control unit **32** exhausts the gas in the closed space via the fluorine trap **37a**. Specifically, the valves EX-V1, EX-V2 are opened while the valves F2-V1, B-V1, L-V1 are kept

closed. Thus, the gas in the laser chamber **10** flows to the exhaust device **37** via the valve EX-V1, and the gas in the pipes **331, 332, 333, 334, 351, 352, 353** flow to the exhaust device **37** via the valve EX-V2. In the exhaust device **37**, the fluorine trap **37a** captures the fluorine gas and the fluorine compounds. The gas having passed through the fluorine trap **37a** is exhausted to the outside of the laser device **1** via the exhaust pump **37b** or the bypass flow path (not shown). When the exhaust pump **37b** is driven, the exhaust can be performed so that the gas pressure in the closed space is lower than the atmospheric pressure.

(64) In **S13**, the gas control unit **32** introduces the helium gas into the closed space. Specifically, while the valves F2-V1, B-V1 are kept closed, the driving of the exhaust pump **37b** is stopped, and then the valves EX-V1, EX-V2 are closed and the valve L-V1 is opened. Thus, the helium gas is introduced from the pipe **351** into the laser chamber **10** via the pipes **352, 353, 331, 332, 333, 334**. The helium gas is also introduced into a part of the pipes **371, 372**. Thus, gas purge of the closed space by the helium gas is performed. After the introduction of the helium gas, the gas control unit **32** closes the valve L-V1.

(65) In **S14**, the gas control unit **32** determines whether or not the value of the counter N has reached a predetermined value Nmax. The predetermined value Nmax is an integer is equal to or larger than 2.

(66) When the value of the counter N has not reached the predetermined value Nmax (**S14: NO**), the gas control unit **32** advances processing to **S15**. In **S15**, the gas control unit **32** updates the value of N by adding **1** to the value of the counter N. After **S15**, the gas control unit **32** returns processing to **S12**. By repeating the gas exhaust and the introduction of the helium gas a plurality of times, fluorine gas in the closed space including the space inside the laser chamber **10** is removed to a concentration allowing exposure to the atmosphere.

(67) When the value of the counter N has reached the predetermined value Nmax (**S14: YES**), the gas control unit **32** advances processing to **S16**.

(68) In **S16**, the gas control unit **32** notifies an external unit of the completion of removing fluorine gas. For example, the gas control unit **32** transmits a signal indicating the completion of removing fluorine gas to the laser control unit **30**. Triggered by the signal, the laser control unit **30** transmits a notification indicating the completion of removing fluorine gas to the FDC system **120**. In addition to the notification to the FDC system **120**, or in place of the notification to the FDC system **120**, the laser control unit **30** causes the user interface **140** to display an indication that removing of the fluorine gas is completed.

(69) In **S17**, the operator causes the closed space to be exposed to the atmosphere to perform component replacement of the laser chamber **10**. For example, when the window **10a** or **10b** is to be replaced, at least a part of the space inside the laser chamber **10** is exposed to the atmosphere. Alternatively, for example, when the laser chamber **10** itself is to be replaced, a part of the pipes **334, 371** and the like connected to the laser chamber **10** is also exposed to the atmosphere. In **S18**, the operator performs component replacement of the laser chamber **10**.

(70) In **S19**, the operator isolates the closed space from the atmosphere and inputs completion of the maintenance operation using the user interface **140**. The laser control unit **30** receives the signal input to the user interface **140** and transmits a signal indicating the completion of the maintenance operation to the gas control unit **32**.

(71) In **S20**, the gas control unit **32** determines whether or not the signal indicating the completion of the maintenance operation has been received. When the signal indicating the completion of the maintenance operation has not been received (**S20: NO**), the gas control unit **32** waits until the signal indicating the completion of the maintenance operation is received. When the signal indicating the completion of the maintenance operation has been received (**S20: YES**), the gas control unit **32** ends the processing of the present flowchart and returns to processing shown in FIG. 2.

(72) 1.2.3.2 Leak Check

(73) FIG. 5 is a flowchart showing the procedure of the leak check in the comparative example. The flowchart of FIG. 5 corresponds to the subroutine of S30 in FIG. 2. In the procedure shown in FIG. 5, steps each surrounded by a solid line frame indicate operation of the gas control unit 32, and steps each surrounded by a broken line frame indicate operation of the operator.

(74) In S32, the gas control unit 32 opens the valve L-V1 and introduces the helium gas into the closed space. At this time, the valves F2-V1, B-V1, EX-V1, EX-V2 are closed.

(75) In S33, the gas control unit 32 determines whether or not the gas pressure in the laser chamber 10 measured by the pressure gauge P has reached a predetermined value. When the gas pressure in the laser chamber 10 has not reached the predetermined value (S33: NO), the gas control unit 32 waits until the gas pressure in the laser chamber 10 reaches the predetermined value. When the gas pressure in the laser chamber 10 has reached the predetermined value (S33: YES), the gas control unit 32 advances processing to S34.

(76) In S34, the gas control unit 32 closes the valve L-V1 and notifies an external unit of the completion of gas filling. For example, the gas control unit 32 transmits a signal indicating completion of gas filling to the laser control unit 30, and the laser control unit 30 transmits a notification indicating completion of gas filling to the FDC system 120 and/or causes the user interface 140 to display an indication that gas filling is completed.

(77) In S35, the operator uses a helium leak detector (not shown) to determine whether or not the helium gas is leaking to the outside of the closed space. For example, when the window 10a or 10b has been replaced, it is checked whether or not the helium gas is leaking around the replaced window. Thereafter, the operator inputs the determination result using the user interface 140. The laser control unit 30 receives the signal input to the user interface 140 and transmits a signal indicating the determination result to the gas control unit 32. Alternatively, a signal indicating the determination result is transmitted from the helium leak detector to the laser control unit 30, and the laser control unit 30 may transmit the signal to the gas control unit 32.

(78) In S36, the gas control unit 32 identifies whether or not the helium gas is leaking to the outside of the closed space.

(79) When the helium gas is leaking to the outside of the closed space (S36: YES), the gas control unit 32 terminates the processing of the present flowchart and returns to processing shown in FIG. 2. At this time, the gas control unit 32 may transmit a signal indicating a device error to the laser control unit 30. The laser control unit 30 transmits a notification indicating the device error to the FDC system 120 and/or causes the user interface 140 to display an indication of the device error.

(80) When the helium gas does not leak to the outside of the closed space (S36: NO), the gas control unit 32 advances processing to S37. In S37, the gas control unit 32 exhausts the gas in the closed space. The exhaust procedure may be similar to that described in S12 of FIG. 4. Thereafter, in S38, the gas control unit 32 introduces the neon-containing gas and the fluorine-containing gas into the closed space including the space inside the laser chamber 10. Specifically, first, the valves EX-V1, EX-V2, F2-V1, L-V1 are closed and the valve B-V1 is opened. Thus, the neon-containing gas is introduced from the pipe 332 into the laser chamber 10. Next, the valves EX-V1, EX-V2, B-V1, L-V1 are closed and the valve F2-V1 is opened. Thus, the fluorine-containing gas is introduced from the pipe 331 into the laser chamber 10.

(81) When the gas pressure and fluorine gas concentration in the laser chamber 10 reaches a desired value, the gas control unit 32 closes the valves EX-V1, EX-V2, F2-V1, B-V1, L-V1 and transmits a signal indicating completion of the operation preparation of the laser device 1 to the laser control unit 30.

(82) Alternatively, in order to stabilize gas composition in the laser chamber 10, the signal indicating completion of the operation preparation of the laser device 1 may be transmitted to the laser control unit 30 after repeating the exhaust in S37 and the gas introduction in S38 a plurality of times.

(83) After S38, the gas control unit 32 terminates the processing of the present flowchart and

returns to processing shown in FIG. 2.

(84) 1.3 Problems of Comparative Example

(85) In the comparative example, the gas purge of the closed space and the leak check are performed using helium gas. However, since helium gas has a limited production area and a limited production amount, there may be a case that procuring helium gas becomes difficult or the price thereof rises. If the procurement of helium gas is stopped, there is a possibility that the maintenance of the laser device **1** cannot be performed and the laser device **1** cannot be operated.

(86) 2. Laser Device that Performs Leak Check Using Neon-Containing Gas

(87) 2.1 Configuration

(88) A first embodiment will be described with reference to FIGS. **6** to **8**.

(89) FIG. **6** schematically shows the configuration of the laser device **1** according to the first embodiment. In the first embodiment, the helium gas cylinder **25** may be omitted. The pipe **351** is configured to introduce third gas containing neon gas and having fluorine gas concentration lower than that of the fluorine-containing gas into the laser chamber **10** and the pipes **331**, **332**. The pipe **351** corresponds to the third pipe in the present disclosure.

(90) The third gas is preferably gas having the same composition as that of the neon-containing gas. The gas having the same composition refers to gas in which both the components and the mixing ratio thereof are the same. More preferably, the first end of the pipe **351** is connected to the neon-containing gas cylinder **24**. A pipe **242** is connected to the neon-containing gas cylinder **24**. The pipe **242** is branched into a pipe **243** and a pipe **244** at a branch portion **24a**. The pipe **243** is connected to the first end of the pipe **332** and the pipe **244** is connected to the first end of the pipe **351**. The pipes **242**, **243**, **244** correspond to the fourth pipe in the present disclosure.

(91) In other respects, the configuration of the first embodiment is similar to the configuration of the comparative example.

(92) 2.2 Operation

(93) The operation of the laser oscillator and the operation of the optical module purge gas supply device **39** in the first embodiment are similar to those described in the above comparative example. The leak check method of the first embodiment is performed in the procedure similar to that described with reference to FIG. **2**. However, a subroutine described below differs between the first embodiment and the comparative example.

(94) 2.2.1 Maintenance Operation

(95) FIG. **7** is a flowchart showing the procedure of the maintenance operation in the first embodiment. Instead of **S13** of FIG. **4**, in **S13a** of FIG. **7**, the gas control unit **32** introduces the neon-containing gas into the closed space.

(96) In other respects, the processing of FIG. **7** is similar to the processing in the comparative example described with reference to FIG. **4**.

(97) 2.2.2 Leak Check

(98) FIG. **8** is a flowchart showing the procedure of the leak check in the first embodiment. Instead of **S32** of FIG. **5**, in **S32a** of FIG. **8**, the gas control unit **32** introduces the neon-containing gas into the closed space. Further, instead of **S35** of FIG. **5**, in **S35a** of FIG. **8**, the operator determines whether or not neon gas is leaking to the outside of the closed space using a leak detector (not shown) having sensitivity to neon gas. For example, when neon gas exceeding a predetermined threshold value is detected, it is determined that neon gas is leaking to the outside of the closed space. The leak detector having sensitivity to neon gas may be, for example, a device that detects neon gas by detecting a difference in thermal conductivity between reference gas and gas to be measured. The leak detector may be configured to output an indication or alarm based on the detected amount of neon gas.

(99) In other respects, the processing of FIG. **8** is similar to the processing in the comparative example described with reference to FIG. **5**.

(100) 2.3 Effect

(101) (1) According to the first embodiment, neon-containing gas containing neon gas is introduced into the closed space (S32a of FIG. 8), and it is determined whether or not neon gas is leaking to the outside of the closed space (S35a of FIG. 8). Thus, since the leak check can be performed without using helium gas, even in a case that helium gas becomes difficult to be procured, it is possible to prevent the operation of the laser device 1 from being stopped due to maintenance impossibility.

(102) (2) According to the first embodiment, prior to exposing the closed space to the atmosphere, gas in the closed space is exhausted via the fluorine trap 37a (S12 of FIG. 7), neon-containing gas is introduced into the closed space (S13a of FIG. 7), and gas in the closed space is exhausted again via the fluorine trap 37a (S12 of FIG. 7). Thus, the inside of the closed space can be gas-purged and the fluorine gas concentration can be reduced without using helium gas. Further, since the gas purge is performed using the same neon-containing gas as the gas used for the leak check, an increase in the number of pipes is suppressed.

(103) (3) According to the first embodiment, a leak detector having sensitivity to neon gas is used to determine whether or not neon gas is leaking to the outside of the closed space (S35a of FIG. 8). Thus, the leak check can be performed immediately after neon-containing gas is introduced into the laser chamber 10.

(104) (4) According to the first embodiment, when it is determined that neon gas is not leaking to the outside of the closed space, gas in the closed space is exhausted (S37 of FIG. 8), and neon-containing gas and fluorine-containing gas are introduced into the closed space (S38 of FIG. 8). Thus, since it is determined in advance that neon gas is not leaking, it is possible to suppress the leakage of fluorine gas when fluorine-containing gas and the like is introduced. Further, when the leak check is performed using neon-containing gas, the number of times of repeating the exhaust in S37 and the gas introduction in S38 can be reduced as compared with the case in which the leak check is performed using helium gas.

(105) (5) According to the first embodiment, the laser device 1 includes the pipe 331 for introducing fluorine-containing gas containing neon gas and fluorine gas into the laser chamber 10, the pipe 332 for introducing neon-containing gas containing neon gas and having the fluorine gas concentration lower than that of the fluorine-containing gas into the laser chamber 10, and the pipe 351 for introducing the third gas containing neon gas and having the fluorine gas concentration lower than that of the fluorine-containing gas into the laser chamber 10 and the pipe 331. Thus, since the leak check can be performed using the third gas containing neon gas and without using helium gas, even in a case that helium gas becomes difficult to be procured, it is possible to prevent the operation of the laser device 1 from being stopped due to maintenance impossibility.

(106) Although the case in which the third gas is introduced via the pipe 351 has been described, the present disclosure is not limited thereto. Even without the pipe 351, the third gas having the same composition as the neon-containing gas may be introduced via the pipe 332.

(107) (6) According to the first embodiment, the pipe 351 introduces the third gas having the same composition as that of the neon-containing gas into the laser chamber 10 and the pipe 331. Thus, since the neon-containing gas constituting the laser medium gas can be diverted as the third gas for the leak check, the number of procured items of gas can be reduced. In addition, the cost of pipe work for the third gas can be reduced.

(108) (7) According to the first embodiment, the pipe 331 is connected to the fluorine-containing gas cylinder 23 storing fluorine-containing gas, and the pipes 332, 351 are connected to the neon-containing gas cylinder 24 storing neon-containing gas.

(109) Further, according to the first embodiment, the pipes 242, 243, 244 including the branch portion 24a are connected to between the pipes 332, 351 and the neon-containing gas cylinder 24. Thus, the gas cylinder connected to the pipes 332, 351 can be used in common. In addition, the pipe for helium gas is not required, and the cost of pipe work can be reduced.

(110) (8) According to the first embodiment, the pipe 351 introduces the third gas into the laser

chamber **10** and the pipes **331**, **332**. Thus, the leak check can be performed not only for the laser chamber **10** but for the pipes **331**, **332** and the like at the same time.

(111) (9) According to the first embodiment, the laser device **1** includes the exhaust device **37** that exhausts gas from the laser chamber **10** and the pipe **331**. Thus, the leak check can be performed for the exhausted laser chamber **10** and pipe **331** into which the third gas is introduced.

(112) (10) According to the first embodiment, the laser device **1** includes the exhaust device **37** that exhausts gas from the laser chamber **10** and the pipes **331**, **332**. Thus, the leak check can be performed for the exhausted laser chamber **10** and pipes **331**, **332** into which the third gas is introduced.

(113) 3. Laser Device that Performs Leak Check Based on Change in Gas Pressure

(114) A second embodiment will be described with reference to FIG. **9**. The configuration of the second embodiment is similar to that of the first embodiment. The operation of the second embodiment is different from that of the first embodiment in a leak check subroutine described below.

(115) FIG. **9** is a flowchart showing the procedure of the leak check in the second embodiment. Instead of **S35a** of FIG. **8**, in **S35b** of FIG. **9**, the gas control unit **32** determines whether or not the neon-containing gas is leaking to the outside of the closed space by detecting the change of the gas pressure in the laser chamber **10** during a certain period **T**.

(116) For example, the gas control unit **32** receives gas pressure **P1** in the laser chamber **10** measured by the pressure gauge **P** from the laser control unit **30**, and further receives gas pressure **P2** when the certain period **T** has elapsed from the laser control unit **30**. Then, the gas control unit **32** calculates the following value ΔP or dP/dT .

$$\Delta P = |P1 - P2|$$

$$dP/dT = |P1 - P2|/T$$

(117) When ΔP is equal to or smaller than a predetermined threshold value or when dP/dT is equal to or smaller than a predetermined threshold value, the gas control unit **32** determines that the neon-containing gas is not leaking to the outside of the closed space. When ΔP is larger than the predetermined threshold value or when dP/dT is larger than the predetermined threshold value, the gas control unit **32** determines that the neon-containing gas is leaking to the outside of the closed space.

(118) In other respects, the operation of the second embodiment is similar to that of the first embodiment.

(119) According to the second embodiment, it is determined whether or not the neon-containing gas is leaking to the outside of the closed space (**S35b** of FIG. **9**) by detecting the change of gas pressure in the closed space. This makes it possible to automate the leak check and to reduce the workload of the operator.

(120) 4. Laser Device that Performs Leak Check of Replacement Laser Chamber

(121) A third embodiment will be described with reference to FIGS. **10** and **11**. The configuration of the third embodiment is similar to that of the first and second embodiments. The third embodiment is different from the first and second embodiments in that the leak check is performed on a replacement laser chamber **10d**.

(122) 4.1 Replacement Laser Chamber

(123) FIG. **10** schematically shows the configuration of the replacement laser chamber **10d**. The replacement laser chamber **10d** is similar to the laser chamber **10** in that the windows **10a**, **10b** are provided thereto and the electrodes **11a**, **11b** are accommodated. When the above-described maintenance operation includes replacement of the laser chamber **10**, the laser chamber **10** is removed from the laser device **1**, and the replacement laser chamber **10d** shown in FIG. **10** is attached to the laser device **1**.

(124) 4.2 Leak Check Procedure

(125) FIG. **11** is a flowchart showing the procedure of the leak check in the third embodiment. The

step surrounded by a broken line frame indicates the operation of a manufacturer of the laser device **1**.

(126) In **S1**, the manufacturer of the laser device **1** encloses neon-containing gas in the replacement laser chamber **10d** in a manufacturing plant of the laser device **1**, and causes a storage device (not shown) to store the gas pressure **P1** in the replacement laser chamber **10d** and the measurement time **T1**. The storage device may be, for example, a storage device attached to the replacement laser chamber **10d**. Alternatively, the storage device may be a storage device accessible from the gas control unit **32** or the laser control unit **30** of the laser device **1** installed in a semiconductor manufacturing plant to which the laser device **1** is shipped.

(127) The replacement laser chamber **10d** is shipped to the semiconductor manufacturing plant and attached to the laser device **1** by laser chamber replacement operation of the laser device **1**.

(128) In **S2**, the gas control unit **32** determines whether or not a signal indicating the completion of the laser chamber replacement operation has been received. When the signal indicating the completion of the laser chamber replacement operation has not been received (**S2: NO**), the gas control unit **32** waits until the signal indicating the completion of the laser chamber replacement operation is received. When the signal indicating the completion of the laser chamber replacement operation has been received (**S2: YES**), the gas control unit **32** advances processing to **S3**.

(129) In **S3**, the gas control unit **32** receives the gas pressure **P2** in the laser chamber **10** measured by the pressure gauge **P** and the measurement time **T2** from the laser control unit **30**. The laser chamber **10** in this case is a replacement laser chamber **10d** newly attached. The gas control unit **32** reads out the gas pressure **P1** and the measurement time **T1** stored in the storage device and calculates the following value ΔP or dP/dT .

$$\Delta P = |P1 - P2|$$

$$dP/dT = |P1 - P2| / (T2 - T1)$$

(130) When ΔP is equal to or smaller than a predetermined threshold value or when dP/dT is equal to or smaller than a predetermined threshold value, the gas control unit **32** determines that the neon-containing gas is not leaking to the outside of the laser chamber **10**. When ΔP is larger than the predetermined threshold value or when dP/dT is larger than the predetermined threshold value, the gas control unit **32** determines that the neon-containing gas is leaking to the outside of the laser chamber **10**.

(131) The processes of **S36** and **S38** are similar to the processes described with reference to FIG. 5. When the neon gas is leaking to the outside of the closed space (**S36: YES**) or after **S38**, the gas control unit **32** terminates processing of the present flowchart.

(132) In other respects, the third embodiment is similar to the first or second embodiment.

(133) 4.3 Effect

(134) According to the third embodiment, the leak check of the replacement laser chamber **10d** is performed (**S3** of FIG. 11), and then the fluorine-containing gas and the like is introduced (**S38** of FIG. 11). Thus, since it is determined in advance that the neon gas is not leaking, it is possible to suppress the leakage of the fluorine gas from the replacement laser chamber **10d** when the fluorine-containing gas or the like is introduced. Further, when the leak check is performed using the neon-containing gas, the number of times of repeating the exhaust in **S37** and the gas introduction in **S38** can be reduced as compared with the case in which the leak check is performed using helium gas.

(135) 5. Others

(136) The description above is intended to be illustrative and the present disclosure is not limited thereto. Therefore, it would be obvious to those skilled in the art that various modifications to the embodiments of the present disclosure would be possible without departing from the spirit and the scope of the appended claims. Further, it would be also obvious to those skilled in the art that embodiments of the present disclosure would be appropriately combined.

(137) The terms used throughout the present specification and the appended claims should be interpreted as non-limiting terms unless clearly described. For example, terms such as “comprise”,

“include”, “have”, and “contain” should not be interpreted to be exclusive of other structural elements. Further, indefinite articles “a/an” described in the present specification and the appended claims should be interpreted to mean “at least one” or “one or more.” Further, “at least one of A, B, and C” should be interpreted to mean any of A, B, C, A+B, A+C, B+C, and A+B+C as well as to include combinations of any thereof and any other than A, B, and C.

Claims

1. A leak check method for a laser device, comprising: exposing a closed space accommodating laser medium gas to the atmosphere; isolating the closed space from the atmosphere after exposing the closed space to the atmosphere; introducing neon-containing gas containing neon gas to the closed space; and determining whether or not the neon gas is leaking to outside of the closed space using a leak detector having sensitivity to neon gas.
2. The leak check method according to claim 1, wherein the closed space includes a space inside a laser chamber.
3. The leak check method according to claim 1, further comprising: before the exposing to the atmosphere, exhausting gas in the closed space via a fluorine trap, introducing the neon-containing gas into the closed space, and exhausting again gas in the closed space via the fluorine trap.
4. The leak check method according to claim 1, wherein the determining is performed based on whether or not neon gas exceeding a threshold value is detected using the leak detector having sensitivity to neon gas.
5. The leak check method according to claim 1, wherein the determining is performed based on change in gas pressure in the closed space.
6. The leak check method according to claim 1, further comprising: when determined that neon gas is not leaking to the outside of the closed space, exhausting gas in the closed space, and introducing the neon-containing gas and fluorine-containing gas into the closed space.
7. A leak check method for a laser device, the laser device including a laser chamber, a first pipe configured to introduce first gas containing neon gas and fluorine gas into the laser chamber, and a second pipe configured to introduce second gas containing neon gas and having fluorine gas concentration lower than that of the first gas into the laser chamber, the leak check method comprising: introducing third gas containing neon gas and having fluorine gas concentration lower than that of the first gas into a closed space including a space inside the laser chamber; and determining whether or not neon gas is leaking to outside of the closed space using a leak detector having sensitivity to neon gas.
8. The leak check method according to claim 7, wherein the third gas is gas having the same composition as the second gas.
9. The leak check method according to claim 8, wherein the first pipe is connected to a first gas supply source storing the first gas, the second pipe is connected to a second gas supply source storing the second gas, and the third gas is introduced into the closed space from the second gas supply source.
10. The leak check method according to claim 7, wherein the laser device further includes a third pipe configured to introduce the third gas into the closed space.
11. The leak check method according to claim 7, wherein the laser device further includes a third pipe configured to introduce the third gas having the same composition as the second gas into the closed space.
12. The leak check method according to claim 11, wherein the first pipe is connected to a first gas supply source storing the first gas, and the second pipe and the third pipe are connected to a second gas supply source storing the second gas.
13. The leak check method according to claim 12, wherein a fourth pipe including a branch portion is connected to between the second pipe and the second gas supply source and to between the third

pipe and the second gas supply source.

14. The leak check method according to claim 7, wherein the determining is performed based on whether or not neon gas exceeding a threshold value is detected using the leak detector having sensitivity to neon gas.

15. A laser device, comprising: a laser chamber; a first pipe connected to a first gas supply source storing first gas containing neon gas and fluorine gas so as to introduce the first gas into the laser chamber; a second pipe connected to a second gas supply source storing second gas containing neon gas and having fluorine gas concentration lower than that of the first gas so as to introduce the second gas into the laser chamber; and a third pipe connected to the second gas supply source so as to introduce the second gas into the laser chamber and the first pipe.

16. The laser device according to claim 15, wherein a fourth pipe including a branch portion is connected to between the second pipe and the second gas supply source and to between the third pipe and the second gas supply source.

17. The laser device according to claim 15, wherein the third pipe introduces the second gas into the laser chamber, the first pipe, and the second pipe.

18. The laser device according to claim 15, further comprising an exhaust device that exhausts gas from the laser chamber and the first pipe.

19. The laser device according to claim 15, further comprising an exhaust device that exhausts gas from the laser chamber, the first pipe, and the second pipe.

20. The laser device according to claim 15, wherein the first pipe includes a first connection portion at which the second pipe is connected to the first pipe and a second connection portion at which the third pipe is connected to the first pipe, the first connection portion and the second connection portion being separate from each other.
